

Particle Physics
Winter Quarter 2024
Physics 130A



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Physics 317

Lecture: M,W,F 9:00-9:50 AM in Roessler 148

Textbooks: Introduction to Elementary Particles (2nd), Griffiths.

Course TA: (No Course TA)

Office Hours: Mulhearn: Monday 10:00-11:00 AM PHY 317

Homework: From 5-10 homework problem sets.

Quizzes: (Quizzes were dropped)

Midterm Exam: Take-Home

Final Exam: March, 21 2024 8:00 AM

Course Description:

Properties and classification of elementary particles and their interactions. Experimental techniques. Conservation laws and symmetries. Strong, electromagnetic, and weak interactions. Introduction to Feynman calculus.

Homework:

There will be 5-10 homework assignments. They will generally be due on Wednesday before lecture at 9 AM by submission to gradescope. You may consult with other students but your submitted work must be your own. Your homework will be graded on effort only.

Exams:

Details on the midterm and final will be discussed later in the quarter.

Grades:

Final grades will be approximately 50% homework, 25% midterm exam, and 25% final

exam. Your worst homework score will be dropped. You must also pass each component of the course (homework, quizzes, and both exams) in order to pass the course. The passing level may be adjusted (lower than 60%) at my discretion. As this is the first time I have taught this course, I may adjust the relative weighting of each contribution to your final score, based on my observation of typical student outcomes.

Course Schedule:

This is a preliminary schedule which will evolve as the course progresses.

Week	Dates	Topics	Reading
1	Jan 8,10,12	Historical Introduction	1.1-1.5
2	Jan 17,19		1.6-1.11
3	Jan 22,24,26	Elementary Particle Dynamics	2.1-2.6
4	Jan 29,31, Feb 2		
5	Feb 5,7,9	Relativistic Kinematics	3.1-3.5
6	Feb 12,14,16		
7	Feb 21,23	Feynman Calculus	6.1-6.3
8	Feb 26,28, Mar 1		
9	Mar 4,6,8	Quantum Electrodynamics	7.1-7.3
10	Mar 11,13,15	Quantum Electrodynamics	7.5-7.8